Recurrent Graph Neural Networks

Logical Characterizations via Modal Logic

Kevin Kunkel & Thomas Mohr

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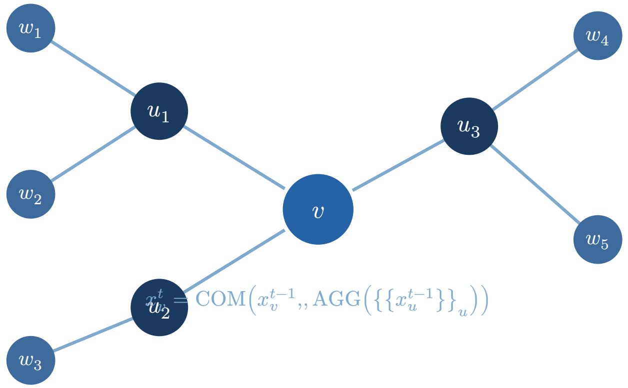
Supervisor: Prof. Carsten Lutz



Motivation

Why Study GNN Expressiveness?

- GNNs are powerful — but **what exactly** can they compute?
- Classic result [1]: constant-iteration GNNs $\equiv$ graded modal logic GML *(relative to first-order definable properties)*
- Missing piece: GNNs that run for an **adaptive** number of steps

Question: What is the expressive power of recurrent GNNs — ones that halt only when a node's state enters an *accepting set*?

Background

Message Passing GNNs

Each node v maintains a feature vector updated in rounds:

$$x_v^{(t)} = \varphi\Big(x_v^{(t-1)},, \bigoplus_{u \in \mathcal{N}(v)} \psi\Big(x_v^{(t-1)}, x_u^{(t-1)}\Big)\Big)$$

- φ – **combination** function (MLP, GRU, ...)
- $\oplus$ – **aggregation** over neighbor multiset (sum, max, ...)
- $\mathcal{N}(v)$ – out-neighbors of v

Standard GNNs run for a fixed N rounds then read out.

Recurrent GNNs run until a termination condition is met.

Definition: Recurrent GNN A recurrent GNN[$\mathbb{R}$] over (Π, d) is a tuple $\mathcal{G} = \left(\mathbb{R}^d, \pi, \delta, F\right)$ with

- **init** $\pi : \mathcal{P}(\Pi) \to \mathbb{R}^d$, **transition** $\delta(x, y) = \mathrm{COM}(x, \mathrm{AGG}(y))$, **accept** $F \subseteq \mathbb{R}^d$

Node v starts at $x_v^0 = \pi(\lambda(v))$ and updates: $x_v^t = \mathrm{COM}(x_v^{t-1},, \mathrm{AGG}(\{\{x_w^{t-1} \mid (v, w) \in E\}\}))$

$\mathcal{G}$ **accepts** (G, v) iff $x_v^t \in F$ for some $t \in \mathbb{N}$.

Replace $\mathbb{R}$ with floating-point numbers $\mathbb{F}$ to get GNN[$\mathbb{F}$].

R-Simple Aggregate-Combine GNNs

The **R-simple** variant uses truncated ReLU and a fixed linear architecture:

$$\mathrm{COM}(x, \mathrm{AGG}(y)) = f\left(x \cdot C + \sum_{u} x_u \cdot A + \boldsymbol{b}\right), \quad f = \mathrm{ReLU}^*$$

where $C, A \in \mathbb{R}^{d \times d}$, $\boldsymbol{b} \in \mathbb{R}^d$, and $\mathrm{ReLU}^*(x) = \min(\max(0, x), 1)$.

> **Remark** Despite their simplicity, R-simple GNNs are as expressive as general GNN[$\mathbb{F}$]s — one of the main theorems.

Modal Logics

Logical Toolkit

GML — Graded Modal Logic

Propositional logic + counting modalities
$\langle k \rangle \varphi$: "$\geq k$ out-neighbors satisfy φ"

Captures constant-iteration GNNs [1]

MSC — Modal Substitution Calculus

GML + rule-based fixpoint substitutions

GMSC adds counting to MSC

ω-**GML** — infinite disjunctions of GML formulas

Expressive power: GML $\subset$ GMSC $\subset$ ω-GML

Main Results

Theorem: Floats [2] The following have the **same expressive power**:

$$\mathrm{GNN}[\mathbb{F}] \quad \equiv \quad \text{R-simple AC-GNN}[\mathbb{F}] \quad \equiv \quad \mathrm{GMSC}$$

- The result is **absolute** — no relativization to a background logic
- Only assumption: $\mathbb{F}$ is a discretely-ordered semiring with decidable equality
- Reduces general GNN[$\mathbb{F}$]s to the simple R-simple form

Theorem: Reals [2]

$$\mathrm{GNN}[\mathbb{R}] \quad \equiv \quad \omega\text{-GML}$$

- Also **absolute** — no background logic required
- ω-GML can define **undecidable** graph properties
- Hence $\mathrm{GNN}[\mathbb{R}]$s are strictly more powerful than $\mathrm{GNN}[\mathbb{F}]$s in general

Theorem 3 — MSO Collapse

Theorem: MSO Collapse [2] For any property $\mathcal{P}$ expressible in MSO:

$$\mathcal{P} \text{ expressible as } \mathrm{GNN}[\mathbb{R}] \Leftrightarrow \mathcal{P} \text{ expressible as } \mathrm{GNN}[\mathbb{F}]$$

Both are further captured by GMSC.

- The real–float gap **vanishes** for MSO-definable properties
- Practical GNNs (float arithmetic) lose **nothing** vs.
  theoretical models (reals) for any MSO-definable target property

Distributed Automata

Counting Message-Passing Automata

Definition: CMPA A **counting message-passing automaton** (Q, q_0, δ, F) runs on graphs: each node updates its state based on its own state and the **multiset** of neighbor states — just like a GNN, but with discrete states.

- **FCMPA**: finite Q **CMPA**: countably infinite Q

The full correspondence:

$$\mathrm{GNN}[\mathbb{F}] \equiv \mathrm{FCMPA} < \mathrm{GNN}[\mathbb{R}] \equiv \mathrm{CMPA}$$

- FCMPAs are the finite-state heart of floating-point GNNs
- CMPAs capture the full power of real-valued recurrent computation on graphs
- Links GNNs to **distributed computing** models

Summary

Results at a Glance

Setting	GNN model	Logic / Automaton
Absolute (floats)	$\mathrm{GNN}[\mathbb{F}]$	$\mathrm{GMSC} \equiv \mathrm{FCMPA}$
Absolute (reals)	$\mathrm{GNN}[\mathbb{R}]$	$\omega\text{-GML} \equiv \mathrm{CMPA}$
Relative to MSO	$\mathrm{GNN}[\mathbb{F}] \equiv \mathrm{GNN}[\mathbb{R}]$	GMSC

Key takeaway: **theory and practice converge** for MSO-definable properties.

Conclusion & Open Questions

What we showed:

- Exact logical characterizations of recurrent GNNs, for both $\mathbb{R}$ and $\mathbb{F}$
- Absolute results — no background logic needed
- Real–float equivalence for all MSO-definable properties

Open questions:

- GNNs with **global readout** — how does the characterization change?
- Complexity of deciding expressibility in GMSC?
- Attention-based architectures (GAT, Transformer) in this framework?

References

References

[1] P. Barceló, E. V. Kostylev, M. Monet, J. Pérez, J. Reutter, and J. P. Silva, "The logical expressiveness of graph neural networks," in *International Conference on Learning Representations (ICLR)*, 2020.

[2] V. Ahvonen, D. Heiman, A. Kuusisto, and C. Lutz, "Logical Characterizations of Recurrent Graph Neural Networks with Reals and Floats," *arXiv preprint arXiv:2405.14606*, 2024.